Technical Briefing: Mechanics and Modeling of Suspended Sediment Transport

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Executive Summary

Suspended sediment transport is a primary mode of sediment movement in fluvial and coastal systems, where particles are maintained within the water column by turbulent mixing rather than remaining in contact with the bed.

This briefing outlines:

- the physical mechanisms of suspension,
- the governing equations for vertical concentration distribution (specifically the Rouse Equation), and
- methodologies for calculating total suspended load transport rates.

Critical Takeaways

- **Balance of Forces**
  Suspension is maintained by a dynamic equilibrium between downward gravitational settling and upward turbulent diffusion.
- **The Rouse Number (Z)**
  This dimensionless parameter serves as the primary indicator of transport mode; $Z < 1$signifies dominant suspended load.
- **Reference Concentration**
  Accurate modeling requires a near-bed boundary condition, often derived using the Einstein relation, which links suspended load to bed-load transport.
- **Integration Requirements**
  Total transport rates are obtained by integrating the product of concentration and velocity profiles across the flow depth. Due to strong gradients near the bed, this process often requires numerical methods.

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1. Physical Mechanisms of Sediment Suspension

1.1 Distinction Between Bed Load and Suspended Load

Sediment transport is categorized into two fundamental modes based on the particle's relationship with the channel bed:

- **Bed Load**
  Particles move by rolling, sliding, or saltation, maintaining intermittent or continuous contact with the bed.
- **Suspended Load**
  Particles are supported by turbulent mixing and move throughout the water column.

Although distinct, a continuum often exists between these modes, particularly in the region immediately above the bed.

1.2 The Mechanism of Suspension

Suspension occurs when upward turbulent mixing balances the downward settling velocity V_s of particles due to gravity.

This balance is described by two competing fluxes:

- **Downward Settling Flux**

$$F_s = V_s C$$

- **Upward Turbulent Diffusion**

$$F_d = -\varepsilon_s \frac{dC}{dy}$$

1.3 Role of Turbulent Coherent Structures

Sediment entrainment is not governed solely by mean flow conditions, but by near-bed turbulent structures, including:

- **Sweeps and Ejections**
  Localized flow motions interacting with the bed
- **Bursting Events**
  Instantaneous turbulent fluctuations that lift particles into the flow, enabling transition to suspension

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2. Mathematical Framework and Governing Equations

2.1 Velocity Decomposition

Flow velocity is decomposed into mean and fluctuating components:

$$u = \bar{u} + u'$$

In steady, uniform open-channel flow:

- Mean vertical velocity is zero ($\bar{v} = 0$)
- Fluctuating components drive vertical sediment mixing

2.2 General and Simplified Transport Equations

The general advection–diffusion equation is:

$$\frac{\partial C}{\partial t} + U \cdot \nabla C = \nabla \cdot \varepsilon_s \nabla C$$

For steady, uniform flow, this simplifies to:

$$V_s C + \varepsilon_s \frac{dC}{dy} = 0$$

which states that downward settling is balanced by upward diffusion.

2.3 Sediment Diffusivity and Velocity Distribution

- **Sediment Diffusivity**

$$\varepsilon_s = \beta_s \varepsilon$$

where $\beta_s$typically ranges from 0.5 to 2.

- **Eddy Viscosity Distribution**

$$\varepsilon = \kappa u_* y \frac{H - y}{H}$$

- **Logarithmic Velocity Profile**

$$u = \frac{u_*}{\kappa} \ln\left(\frac{30y}{k_s}\right)$$

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3. The Rouse Concentration Distribution

The Rouse equation relates concentration at elevation $y$to a reference concentration $C_a$at elevation a.

3.1 The Rouse Equation

$$\frac{C}{C_a} = \left[\left(\frac{H-y}{y}\right)\left(\frac{a}{H-a}\right)\right]^Z$$

3.2 Interpretation of the Rouse Number (Z)

$$Z = \frac{V_s}{\kappa u_*}$$

Rouse Number Z	Transport Mode
$Z > 2.5$	Bed load dominant
$1 < Z < 2.5$	Mixed load
$Z < 1$	Suspended load dominant

- Small Z: strong suspension, more uniform distribution
- Large Z: rapid settling, concentration near the bed

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4. Quantifying Suspended Load Transport

4.1 Reference Concentration (C_a)

The Rouse profile requires a boundary condition near the bed:

- **Reference Elevation**

$$a \approx 2D$$

- **Einstein Relation**

$$C_a = \frac{q_{sb}}{11.6\, u'_* a}$$

where $u'_*$is the grain shear velocity.

4.2 Suspended Load Transport Rate (q_{ss})

The total suspended load per unit width is:

$$q_{ss} = \int_a^H C(y)\, u(y)\, dy$$

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5. Computational Strategies and Numerical Integration

Calculation of $q_{ss}$is challenging because the integrand varies rapidly near the bed.

- **Logarithmic Transformation**

$$x = \ln\left(\frac{y}{a}\right)$$

clusters computational points near the bed

- **Numerical Methods**
  - Trapezoidal Rule commonly used
- **Einstein Integrals (I_1, I_2)**
  - Pre-calculated values used to simplify integration
  - Based on Rouse number and $A = a/H$

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6. Alternative Profiles and Limitations

6.1 Alternative Vertical Profiles

- **Rouse Profile**
  Standard for teaching and modeling
- **Lane–Kalinske**
  Exponential form assuming constant diffusivity (simpler but less accurate)
- **van Rijn**
  Two-layer model with improved near-bed accuracy

6.2 Limitations of Classical Theory

Classical theory relies on several simplifying assumptions:

- **Equilibrium Transport**
  Assumes steady state, while natural systems often exhibit non-equilibrium conditions
- **Clear-Water Turbulence**
  Assumes sediment does not affect flow, though high concentrations can reduce turbulence
- **Constant Parameters**
  Assumes constant settling velocity and diffusivity, while both may vary in reality